Dynamic FDTD Modeling of Ultrafast Laser–Hybrid Grating Interaction and Damage Mechanisms

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INTRODUCTION

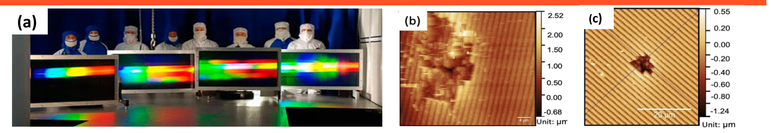


Figure 1: Gratings developed at Lawrence Livermore National Laboratory. (a) Giant diffraction grating for petawatt-class ultrafast laser [1]. (b,c) Typical laser-induced damage observed in the multilayer dielectric gratings [2].

- Future technologies like **laser-based accelerators** and **inertial fusion drivers** require high energy high power intense lasers [3].
- **Gratings and Mirrors** are identified as the weakest parts in the Chirped Pulse Amplification system [4]. It is crucial to increase their laser-induced damage threshold (LIDT).
- A quantitative framework for analyzing damage precursors in complex optical coatings and supports the design and optimization of high-LIDT optical components for **petawatt-class laser systems** such as OMEGA EP [5], EP-OPAL, and NIF ARC [6] is demanding.

THEORY

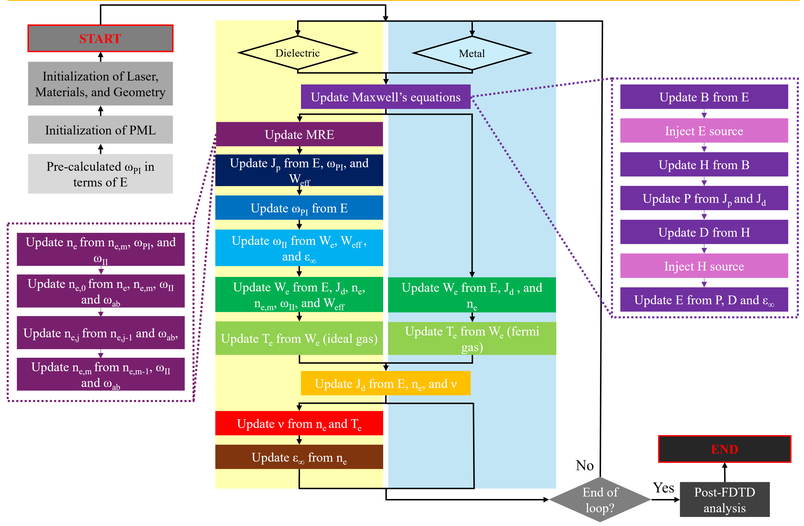
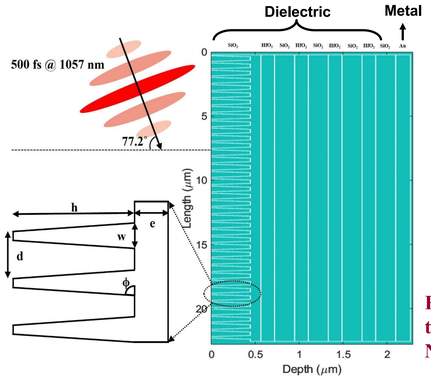


Figure 2: Illustration of the framework. The initialization procedures are not present in full, and only major transient parameters are shown in the workflow.

- Our custom-built **finite-difference time-domain (FDTD)** framework is used to model hybrid coating interactions under ultrafast laser irradiation [7,8].
- **Keldysh's strong-field photoionization** and **impact ionization** are included to describe free-electron generation [9,10].
- To capture the non-thermal nature of impact ionization through dynamic electron excitation, **multiple-rate equations (MRE)** are employed instead of a single-rate model[11].
- The **Drude model** is used to represent the transient optical response of both dielectric and metallic layers.
- **Ohmic heating** is incorporated to account for laser energy absorption by the free-electron plasma.
- **Perfectly matched layer (PML)** and **Total-field scattered-field (TFSF)** are implemented to reduce the boundary reflection and enable long pulses injection without prescribing the entire pulse in the spatial domain [12,13].

SIMULATION SETUP



The Stack Design:
Sub | Au | $(SiO_2 | HfO_2)^4$ | SiO_2 | Air

The Grating Design:
h = 456 nm, d = 562 nm,
w = 257 nm, e = 124 nm, ϕ = 78.2°

- The simulation box is 22.8 μm by 3.6 μm, with a resolution of 12 nm.
- The simulated fluences range from 2.5 to 3.7 J/cm².

Figure 3: The set-up of the pulse and target schematic illustration. Design by Neauport[14,15].

Table 1: Material properties used in the simulations.

Properties	m_e	m_i	W_0(eV)	n + i κ	Mass Density (g/cc)
a-SiO_2	0.6	8	9	1.45	2.20
a-HfO_2	1.09	1.12	5.7	2.06	9.68
Au	1	1	\	0.17 + 7.05 i	19.32

RESULTS

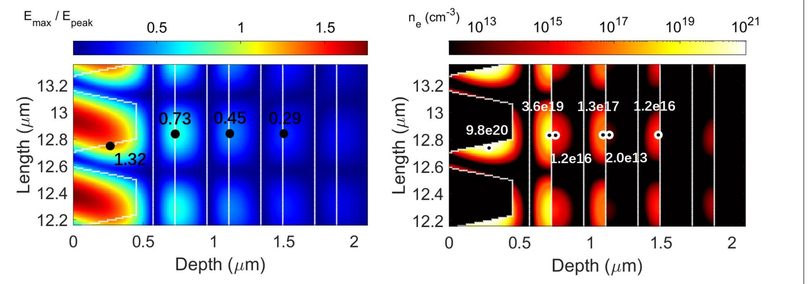


Figure 4: (a) The electric-field distribution and (b) the corresponding electron density distribution in the dielectric region at 3.2 J/cm².

- Damage is predicted to initiate at the **pillar sidewalls**, followed by damage in the **first HfO_2 layer** as the electron density approaches the critical density.
- Experiments also show that damage initiates at the grating surface [14,15].

LIDT Criteria

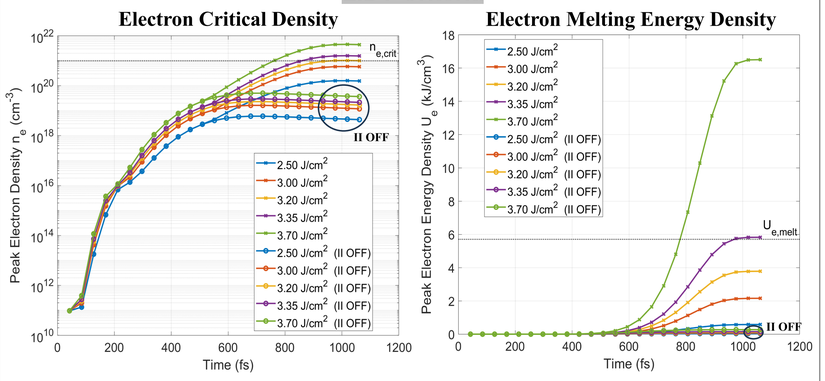


Figure 5: Temporal evolution of the peak electron density and electron energy density in the top SiO_2 grating pillar up to 1.06 ps at different fluences ranging from 2.5 to 3.7 J/cm² , comparing simulations with impact ionization enabled and disabled, with the dashed line indicating the critical plasma density and melting energy density threshold.

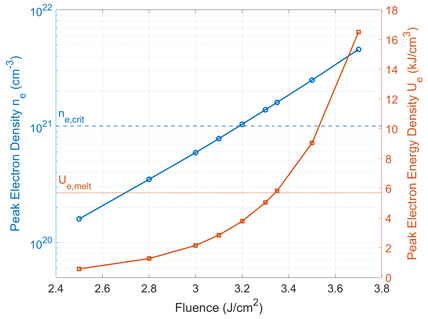


Figure 6: LIDT predicted using the critical electron density and melting electron energy density criteria.

- The plasma critical electron density at 1057 nm is about 10^{21} cm^{-3}, and melting energy density is about 5.7 kJ cm^{-3} [16,17].
- Based on the plasma critical density and melting energy density criteria, LIDT is predicted to be between 3.2-3.35 J/cm², which is in good agreement of reported 3.24 J/cm² under 100-on-1 ablation test [15].

Empirical Power law – LIDT vs. Duration

- Predicted LIDT at different duration for hybrid grating design from Xu et al. [18] and Guan et al. [19] are also presented and compared with Neauport et al. [15].
- The power-law coefficient ranges from 0.37 to 0.63, consistent with previous reports showing that the scaling exponent deviates from 0.5 [20] and typically are 0.3–0.6.

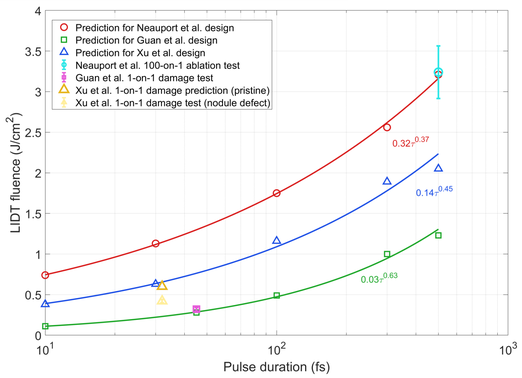


Figure 7: LIDT of the three designs as a function of pulse duration, predicted by the model and fitted with power-law scaling. Experimentally measured LIDT values are also indicated for comparison.

CONCLUSION

1. Extended **LIDT prediction framework** for **hybrid dielectric coatings** with excellent agreement with experiments [14,15,18,19].
2. Comparison of **critical density** and **energy density criteria** for LIDT prediction.
3. **TFST implementation** enables simulations from **10 fs to 1 ps** and reveals ultrafast LIDT power-law scaling, with scaling factors slightly higher than bulk/mirror in the fs regime but generally **< 0.5**.
4. Our framework helps guide the development of **next generation high power laser optics**.

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